

Exercise 01

PART A

For each of the following questions there are four choices marked A, B, C and D. Only one of them is correct. You should decide on the correct choice and write down its mark on the answer sheet.

1. Which of the following is not true in database systems?
A. DBMS provides DDL to define database. B. Database managed by DBMS.
C. Share is not completely. D. Any person can access any data in the database.
2. The relationship among DB, DB application and DBS is _____.
A. A DB system contains multiple DB applications, which access DB
B. A DB application contains only one DB system, which accesses DB
C. A DB contains multiple DB applications, which access DB system.
D. A DB application contains multiple DBs, which access DB system
3. The most appropriate method to manage data with high data sharing and low data redundancy is _____.
A. file-based approach B. entity-relationship approach
C. data-driven approach D. database approach
4. In the following properties, the one that is not belong to database approach is _____.
A. high data sharing B. improved data integrity
C. high data redundancy D. improved data independence
5. In database systems, data independence means that _____.
A. the immunity of application programs to changes of the data value
B. the immunity of application programs to changes of the data amounts
C. the immunity of application programs to changes of the data structure
D. the immunity of the data storage structure to changes of the data storage approach
6. The person who is responsible for the physical realization of the database is referred to as _____.
A. DBA B. EMBA C. MBA D. DA [Database Analyst]
7. _____ contains the global logical representation of the entire database.
A. Conceptual schema B. Internal schema
C. External schema D. Physical schema
8. It is _____ that ensures the data independence in database system.
A. the schema and the external schema
B. the schema and the internal schema
C. the two mappings between the three level of schemas
D. the three level of schemas
9. It is _____ that ensures the logical data independence in database system.
A. the mappings between conceptual schema and external schemas
B. the mapping between conceptual schema and internal schema
C. the conceptual schema
D. the three level of schemas
10. The description of the database logical structure that can be seen by both users and application programs is called _____.
A. conceptual schema B. internal schema

- C. external schema D. physical schema
11. A data model comprises three components, they are _____.
 - A. the external schema, the conceptual schema and the internal schema
 - B. the relations, the tuples and the primary keys
 - C. the structural part, the manipulative part and the integrity rules
 - D. the meaning of model, the operational rules and the implementation language
12. The data in a hierarchical data model is of _____.
 - A. a tree structure B. a directed graph structure
 - C. a linear structure D. a tabular structure
13. The data in a network data model is of _____.
 - A. a tree structure B. a directed graph structure
 - C. a linear structure D. a tabular structure
14. A relational data model is the most important model that is composed of _____.
 - A. the relational data structure, the relational operations and the integrity constraints
 - B. the relations, the tuples and the primary keys
 - C. the entity integrity, the referential integrity and the user-defined integrity
 - D. the external schema, the conceptual schema and the internal schema
15. If we query in a database based on a _____, we need only to express *what* and not *how*.
 - A. hierarchical data model B. network data model
 - C. relational data model D. physical data model
16. Which of the following is true in database systems?
 - A. DBMS runs concurrently with OS B. DBMS and OS call each other.
 - C. OS calls DBMS D. DBMS calls OS.
17. The core of the information processing based on database systems is _____.
 - A. the disk B. program C. the data D. the memory
18. In database system, the conversion between logical data and physical data can be done by _____.
 - A. application programs B. the OS C. the DBMS D. I/O facilities
19. The three-level architecture of database systems is the three-level abstractions for _____.
 - A. memories B. data C. programs D. disks
20. It is just because _____ that there is data independence in database systems.
 - A. the disk-based storage B. the three-level schemas
 - C. the data access by calling OS D. the visualized programming language
21. In database systems, the relation between 'data independence' and 'data relationship' is that _____.
 - A. they are irrelevant concepts B. they are simultaneously true or false
 - C. the former infers the latter D. they can infer each other
22. In database system, 'data independence' means that _____.
 - A. this data is independent on that data
 - B. application programs are independent on the database structure
 - C. disk is independent on memory
 - D. data is independent on disk

PART B

Fill in blanks with proper words or phrases.

1. Database refers to a (a) collection of (b) data, and a (c) of this data, designed to meet the information needs of an organization.
2. The three-level schema architecture of database includes: (a), (b), (c), where there may be multiple (d) in a database, but there are only one (e) and (f) in a database.
3. Roles in the database environment include: (a), (b), (c), (d) and (e).
4. Logical data independence refers to the immunity of (a) to changes in (b). That is: changes in (c) do not require to modify (d) and (e). This is guaranteed by (f) mappings.
5. Physical data independence refers to the immunity of (a) to changes in (b). That is: changes in (c) do not require to modify (d). This is guaranteed by (e) mapping.
6. Data model refers to an integrated collection of concepts for describing (a), (b), and (c). A data model comprises three components: (d), (e) and (f).

PART C

Discuss each of the following terms.

- (a) data
- (b) database
- (c) database management system
- (d) database system
- (e) data independence

PART D

Answer the following questions.

1. Discuss the concept of data independence and explain its importance in a database environment.
2. What is a data model? Describe its components.